SOFTWARE ENGINEERING

Alumni Mentorship & Mock Interview Platform

LAB-1

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DATE-20/8/26

SECTION-AIML-D

PROBLEM STATEMENT-6

Deliverable 1: Requirements Table

FUNCTIONAL REQUIREMENTS (5RFS)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall recommend alumni mentors to students based on domain matching (e.g., Embedded Systems, Cloud Computing, AI/ML) and facilitate one-click session booking.	High	**Pass:** Mentorship session is added to both calendars with meeting link; mentor profile shows domain tags matching student interest. **Fail:** Double-booking occurs; recommended	Core platform functionality — without intelligent matching, the platform fails its primary purpose.

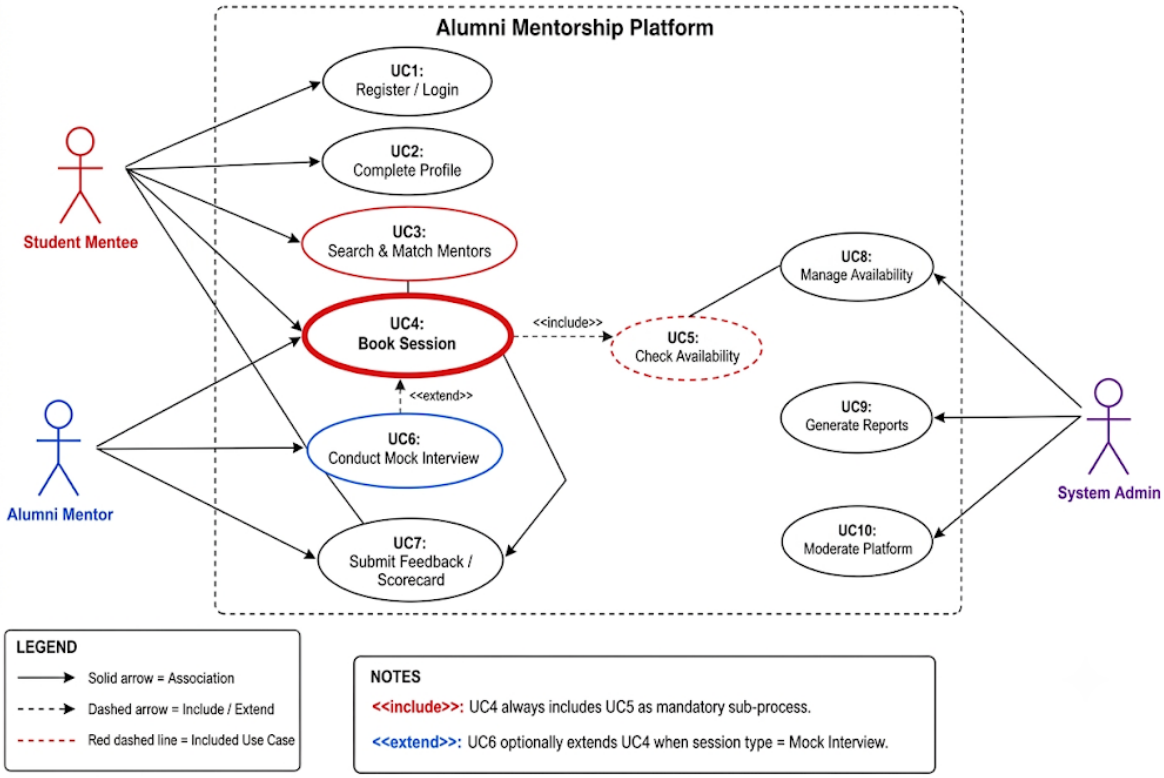
				mentor has zero domain overlap.	
FR-002	Functional	The system shall display real-time mentor availability and allow students to book 30/60-minute slots with automatic calendar sync.	High	**Pass:** Booking reflects immediately on both dashboards; no overlapping slots accepted; ICS file generated. **Fail:** Student books an already-taken slot; calendar sync delay exceeds 5 minutes.	Ensures reliable scheduling and prevents conflicts, critical for trust between mentee and mentor.
FR-003	Functional	The system shall generate a unique video meeting link for every confirmed session and provide in-session chat and screen-share capabilities.	High	**Pass:** Link is active 10 min before session; both users join without external app dependency. **Fail:** Link invalid at session time; audio/video quality degrades below 480p.	Remote mentorship depends entirely on seamless virtual meeting infrastructure.
FR-004	Functional	The system shall provide a structured post-session scorecard for mock interviews covering Technical, Communication, Problem-Solving, and Domain Knowledge with 1–5 ratings and textual notes.	Medium	**Pass:** Mentor submits scorecard within 24h; student views aggregated feedback with radar chart; data exportable as PDF. **Fail:** Scorecard fields missing; student cannot view feedback after deadline.	Structured feedback is the key differentiator for mock interviews and drives student improvement.
FR-005	Functional	The system shall allow mentors to define recurring weekly slots, block specific dates, set maximum sessions per week, and receive booking notifications with accept/decline options.	Medium	**Pass:** Mentor updates availability reflected in student search within 1 min; auto-decline if max sessions reached;	Gives mentors control over their time and prevents burnout by enforcing session caps.

				notification delivered via email + in-app. **Fail:** Outdated slots still appear bookable; mentor receives no notification.	

NON-FUNCTIONAL REQUIREMENTS

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional (Performance)	The system shall complete any booking flow (search to confirmation) in under 5 seconds and process scorecard submissions with <200ms latency under 1,000 concurrent users.	High	**Pass:** Load testing confirms <5s booking latency and <200ms API response; no degradation at peak load. **Fail:** Booking takes >10s; timeout errors during peak usage.	Fast response ensures smooth user experience during high-traffic periods like placement seasons.
NFR-002	Nonfunctional (Security & Privacy)	All user profile data and feedback scorecards shall adhere to privacy guidelines (GDPR/FERPA) with AES-256 encryption at rest and in transit, and support soft-deletion within 24 hours.	High	**Pass:** Penetration test reveals no PII leakage; soft-deletion audit trail maintained; encryption verified on all data stores. **Fail:** Unencrypted PII found in logs; hard-deletion occurs without recovery window.	Protects sensitive student and alumni data and ensures regulatory compliance.

Deliverable 2: UML Use-Case Diagram



Deliverable 3: Use-Case Flow Specification

Field	Value
Use-Case ID	UC-04
Use-Case Name	Book Mentorship Session
Primary Actor	Student Mentee
Secondary Actors	Alumni Mentor, System (Calendar Service), Payment Gateway
Description	The Student Mentee selects a matched mentor, views available time slots, chooses a session type (Mentorship or Mock Interview), and confirms the booking. The system validates availability, reserves the slot, creates calendar entries, generates a video meeting link, and notifies both parties.

Trigger	Student clicks the "Book Session" button from a mentor profile or search result.

Preconditions:

1. Student Mentee is authenticated and has completed their profile (UC-02).
2. Alumni Mentor has published at least one future availability slot (UC-08).
3. Mentor's weekly session cap has not been reached.
4. Student has no conflicting booking at the selected time slot.
5. Mentor profile includes at least one domain tag that matches the student's interest area.

Postconditions (Success):

- Session record created with status CONFIRMED.
- Slot deducted from mentor's weekly quota.
- Calendar events synced to both parties.
- Video meeting link generated.
- Confirmation notifications sent.
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Postconditions (Failure):

- Booking rolled back; no session record created.
- Error shown with alternative suggestions.
- No calendar entries or notifications generated.

Main Success Scenario

Step	Actor	Action
1	Student	Clicks "Book Session" on mentor's profile.
2	System	Displays available slots for next 14 days.
3	Student	Selects date, time slot, and session type.
4	System	**«include»** Invokes UC-05: Check Availability.
5	System	Validates no conflicts; cap not exceeded.
6	Student	Enters agenda; clicks "Confirm Booking".
7	System	Creates session record CONFIRMED; reserves slot.
8	System	Generates encrypted video meeting link.
9	System	Syncs calendars via ICS + API push.
10	System	Sends confirmation email + in-app notification.
11	System	Displays success page with meeting link.
12	Use Case	Ends successfully.

Alternate Flow A1 — Slot Becomes Unavailable During Booking

Step	Actor	Action
4a	Student	UC-05 returns SLOT_TAKEN.
4a.1	System	Shows error + 3 nearest alternatives.
4a.2	System	Selects alternative or cancels.
4a.3	System	Resumes at Step 5 if selected; else ends.

Alternate Flow A2 — Mock Interview Session Selected

Step	Actor	Action
3a	Student	Selects "Mock Interview".
3a.1	System	**«extend»** Activates UC-06; pre-loads scorecard.
3a.2	System	Adds 10-min buffer.
3a.3	System	Includes "Practice Problem Set" link.
3a.4	System	Resumes Main Flow at Step 4.

Alternate Flow A3 — Mentor Weekly Cap Reached

Step	Actor	Action
5a	Student	Mentor cap reached after concurrent booking.
5a.1	System	Slot marked "Unavailable"; waitlist offered.
5a.2	System	Joins waitlist or picks different mentor/time.
5a.3	System	If waitlist creates WAITLISTED record; auto-notifies if slot opens within 48h.

Exception Flow E1 — Calendar Sync Failure

Step	Actor	Action
9a	Student	Calendar API returns 5xx or timeout.
9a.1	System	Logs failure; retries async 3× over 2 min.
9a.2	System	Booking stays CONFIRMED ; shows warning.

9a.3	System	Emails ICS fallback to both parties.

Business Rules

Rule ID	Statement
BR-001	A student cannot have more than 3 pending future sessions with the same mentor.
BR-002	Cancellations within 2h of session start incur a "Late Cancel" flag.
BR-003	Mock Interview sessions trigger scorecard reminder to mentor 30 min after session ends.